

SAMPSON COUNTY AP, NC, USA

WMO: 722073

Lat: 34.976N

Lon: 78.364W

Elev: 148

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03727

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.7	26.7	5.0	7.2	28.4	9.8	9.2	34.3	19.0	52.1	16.8	50.4	3.1	10	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	95.2	77.0	92.9	76.3	90.7	75.5	80.0	89.2	79.0	87.9	78.1	86.4	5.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.8	145.4	83.0	76.9	140.8	82.3	75.9	136.1	81.5	43.5	89.2	42.6	88.2	41.6	86.6	93.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	12.5	10.8	DB	16.6	99.1	4.1	2.7	13.6	101.1	11.2	102.7	8.8	104.2	5.8	106.2
			WB	14.9	82.3	4.0	3.6	12.0	84.9	9.6	87.0	7.3	89.0	4.4	91.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	44.6	47.5	53.2	62.8	70.6	77.9	81.0	79.7	74.4	64.1	52.9	49.5
	DBStd	14.91	10.68	9.70	9.50	7.82	6.34	4.62	3.35	3.77	5.24	8.35	8.48	9.12
	HDD50	663	235	151	79	6	0	0	0	0	0	4	63	125
	HDD65	2646	633	496	381	132	25	0	0	0	3	120	370	486
	CDD50	5505	68	80	178	389	637	838	961	919	733	441	151	110
	CDD65	2013	2	5	14	65	198	388	496	454	286	91	8	6
	CDH74	18064	3	33	148	622	1715	3643	4895	4099	2185	653	51	17
	CDH80	7370	0	1	17	148	590	1634	2267	1756	807	149	1	0
Wind	WSAvg	3.6	4.1	4.6	4.7	4.7	3.6	3.1	2.8	2.3	3.1	3.2	3.5	3.5
Precipitation	PrecAvg	54.4	4.0	3.3	4.1	3.5	3.9	5.1	6.1	6.8	6.5	3.6	3.4	3.2
	PrecMax	68.3	7.1	8.7	6.6	6.5	6.4	14.4	12.7	16.2	21.3	7.3	8.8	6.8
	PrecMin	35.9	0.9	1.3	1.0	0.6	1.6	1.4	2.7	2.6	0.3	0.2	0.5	0.9
	PrecStd	8.1	1.8	1.8	1.7	1.6	1.4	2.7	2.6	3.3	4.7	2.3	1.9	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.4	78.9	82.2	87.9	94.4	98.0	98.9	97.3	92.9	88.6	79.1	76.8
		MCWB	65.0	65.6	63.2	67.2	75.0	74.8	76.9	77.3	73.7	74.8	65.6	69.0
	2%	DB	70.3	73.0	78.7	84.0	89.6	94.0	95.1	93.4	90.3	83.8	75.0	72.4
		MCWB	62.8	62.8	64.0	67.1	71.5	75.8	78.0	77.3	73.9	71.4	64.5	66.1
	5%	DB	66.1	68.5	74.0	81.3	86.3	91.3	92.8	90.9	88.1	81.0	71.8	69.0
		MCWB	60.7	59.8	61.2	64.6	71.0	74.8	77.2	76.7	73.9	69.9	63.6	64.8
	10%	DB	62.5	64.0	70.0	77.4	82.9	89.2	90.6	89.0	85.0	77.4	67.9	65.3
		MCWB	56.8	56.9	59.2	64.6	70.0	74.2	76.5	76.2	73.0	68.2	62.7	61.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.9	68.3	68.2	74.1	77.2	80.1	81.3	81.4	79.2	77.6	70.7	71.6
		MCDB	70.8	73.8	76.0	79.6	83.0	90.4	91.3	88.9	87.0	83.0	73.8	74.3
	2%	WB	64.9	65.8	66.2	71.2	74.9	78.5	80.1	79.9	77.5	75.5	68.2	68.1
		MCDB	68.4	71.1	73.6	77.4	83.5	88.2	89.8	88.3	84.9	80.0	71.4	70.2
	5%	WB	61.3	62.5	64.0	68.8	73.3	77.6	79.2	78.8	76.2	73.7	65.8	65.1
		MCDB	65.5	66.5	70.3	75.3	81.4	86.5	88.5	86.9	82.9	77.9	69.7	67.8
	10%	WB	57.8	58.3	61.8	66.7	72.0	76.4	78.2	77.8	74.9	71.5	62.9	62.0
		MCDB	61.7	63.6	68.5	73.8	79.5	84.4	86.9	85.2	80.8	75.2	67.0	65.0
Mean Daily Temperature Range	5% DB	MDBR	19.3	20.6	21.9	22.9	21.2	19.8	18.7	17.6	18.2	20.5	21.4	19.2
		MCDBR	23.4	25.1	26.8	27.0	24.6	22.9	21.3	20.0	21.5	21.7	23.3	20.5
		MCWBR	16.8	16.0	13.7	11.8	9.4	7.4	6.6	6.5	7.8	10.3	13.9	14.6
	5% WB	MCDBR	20.4	22.4	21.9	20.7	20.0	19.6	18.9	17.9	17.5	16.1	18.7	18.6
		MCWBR	17.1	16.8	13.8	11.1	8.7	7.6	7.1	6.6	7.4	8.6	14.2	15.3
Clear-Sky Solar Irradiance	taub	0.320	0.333	0.366	0.404	0.462	0.500	0.540	0.520	0.428	0.382	0.341	0.320	
	taud	2.486	2.443	2.382	2.274	2.153	2.092	1.997	2.047	2.301	2.401	2.473	2.521	
	Ebn at Noon	281	289	287	280	264	253	241	244	264	267	270	274	
	Edh at Noon	27	31	36	42	48	51	56	52	39	32	27	25	
All-Sky Solar Radiation	RadAvg	845	1058	1397	1754	1922	2019	1926	1728	1459	1205	929	741	
	RadStd	53	111	103	115	144	127	118	94	114	146	73	60	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+286	+205

Nomenclature: See separate page